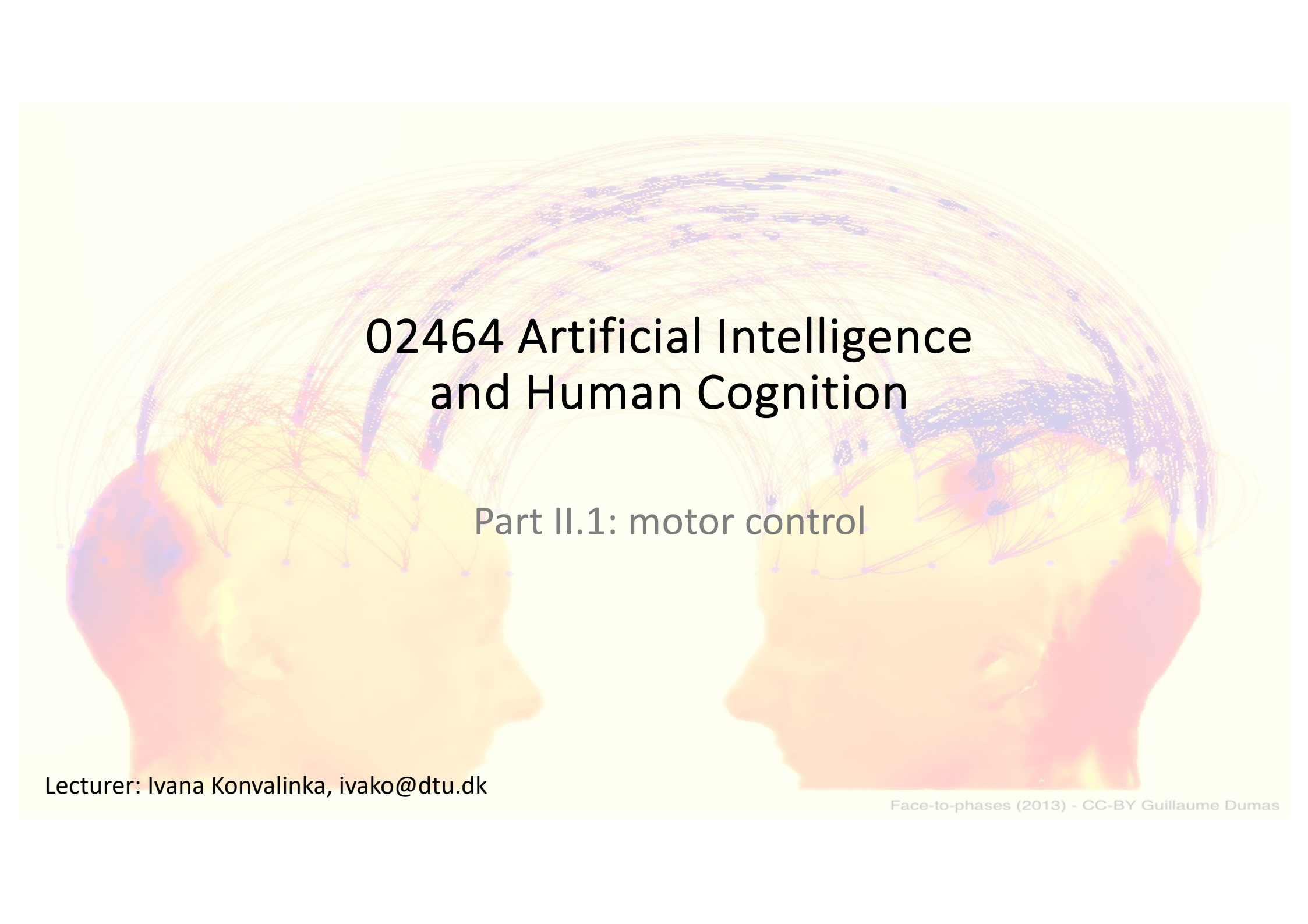




02464 Artificial Intelligence and Human Cognition

Part II.1: motor control

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Face-to-phases (2013) - CC-BY Guillaume Dumas

Overview of 2nd half of the course

- Motor cognition (week 8)
 - Lecture + In-class exercises
- Social cognition (weeks 9 & 10)
 - 2 Lectures + mini-project on multivariate decoding of interactive roles from EEG data
 - Mini-project can be submitted for evaluation (only as a Jupyter notebook!) by Nov. 12th (**you can choose to submit social cognition OR embodied cognition project**)
- Embodied cognition (weeks 1, 11-12)
 - 2 Lectures + HRV/inter-subject correlation mini-project
 - Mini-project can be submitted for evaluation (only as a Jupyter notebook!) by Nov. 26th

Why am I teaching this? Or rather, why are you learning this?

- Our minds are embodied – the body, our movements, our interactions with the environment all shape our cognition, i.e., how we think and perceive the world
- We cannot interact with the world without our body – how we move guides how we interact with the environment and how we socially interact with other people
- Can disembodied AI (e.g., chatGPT) ever approach human-like cognition?
Why or why not?

Motor control learning objectives

- You should be able to:
 - explain why motor control is difficult (for humans and AI)
 - uncertainty, sensory/motor noise, time delays, etc.
 - describe the concept of internal models, and differentiate between forward and inverse models in motor control
 - explain what forward models are used for, what their input/output is, and give an example of their use in human motor control
 - explain inverse models, including the control diagram
 - describe and apply Bayes rule in action
 - describe and apply decision theory in motor control

Why do we (and other animals) have a brain?



- To produce adaptable and complex movements: our only way to interact with the world

Chess: AI vs. human cognition

Task:
Choose which piece to
move where

high-level task



Task:
Physically manipulate
each piece and move it
around the board

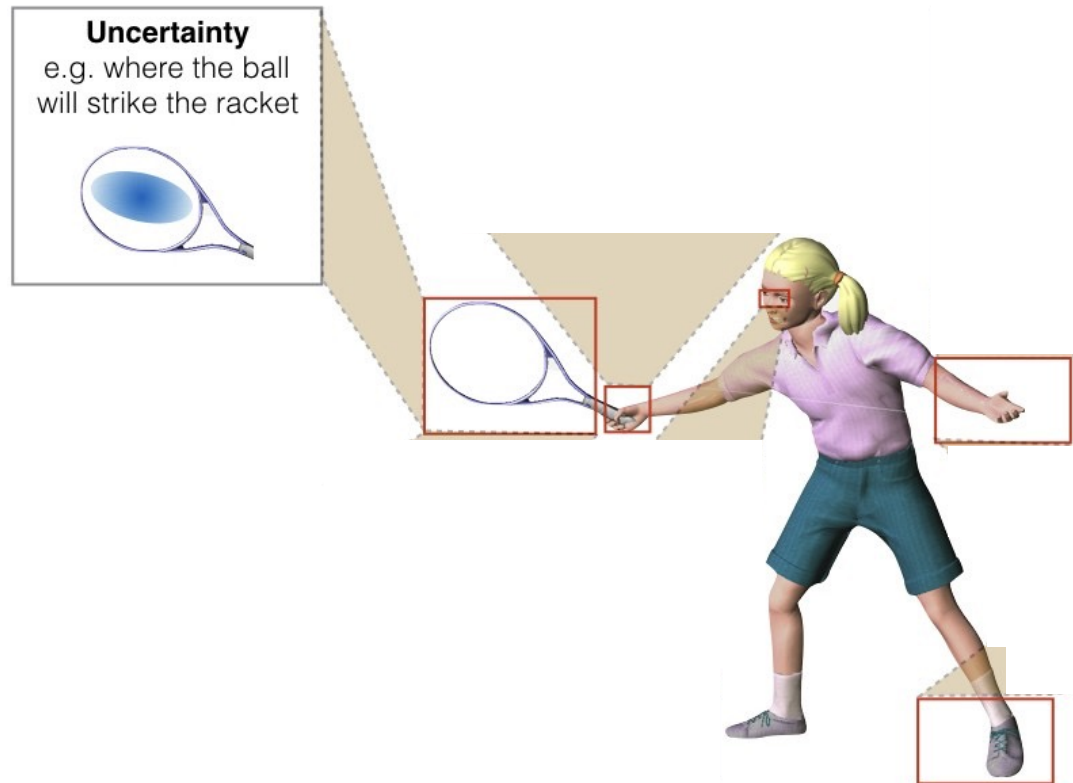
low-level task



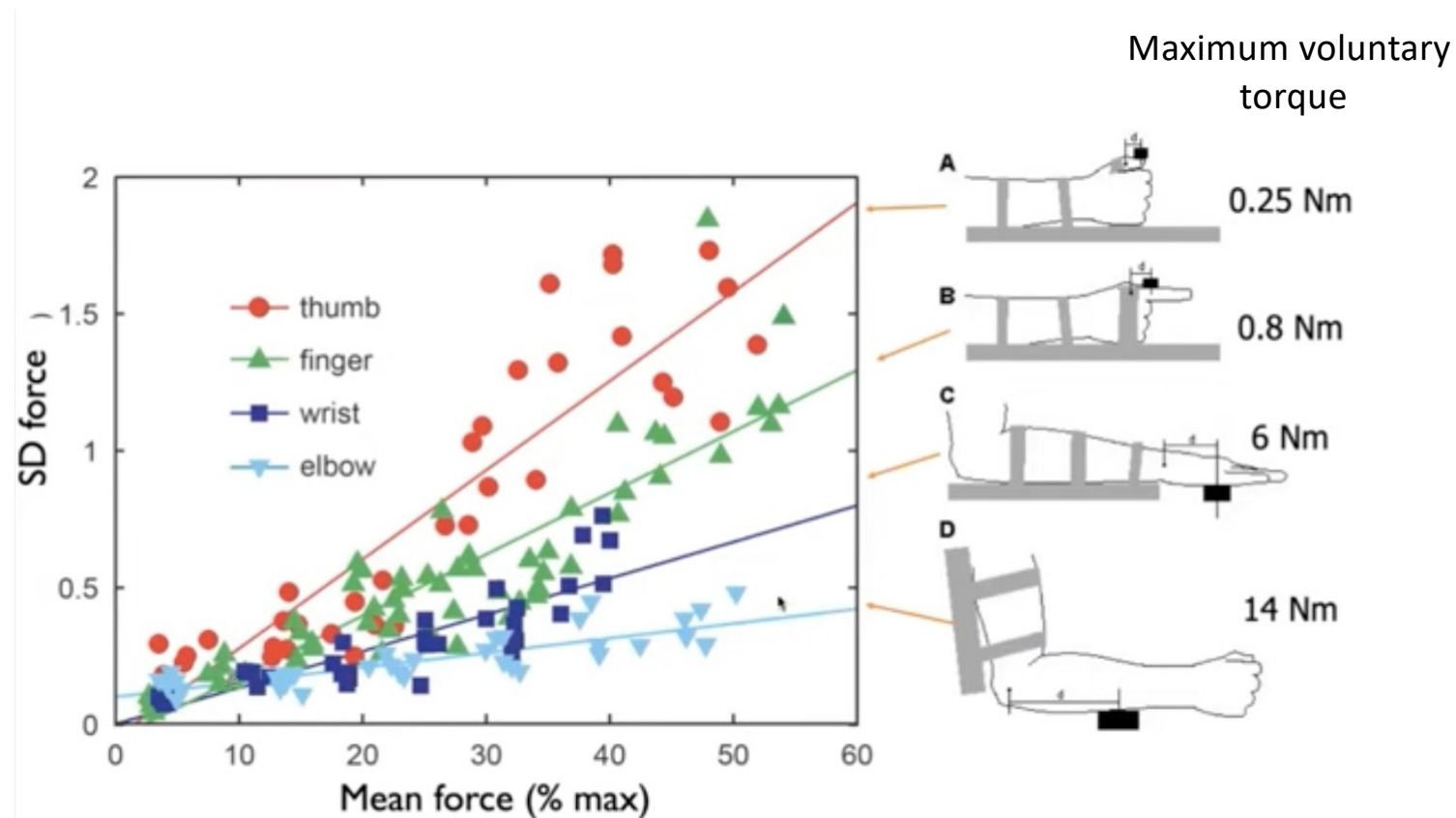
High vs. Low level processes in cognition

- High level:
 - overall goal or intention
 - symbolic or abstract level
 - e.g., cognitive abilities like planning and reasoning, require generalization, may require conscious awareness
 - implementation in AI systems often does not resemble neuronal structures
- Low level:
 - mechanics such as joint rotations and muscle movements, visual processing
 - details of sensory processing
 - computer vision benefits from detailed understanding of the cortical visual system

Why is motor control so difficult?

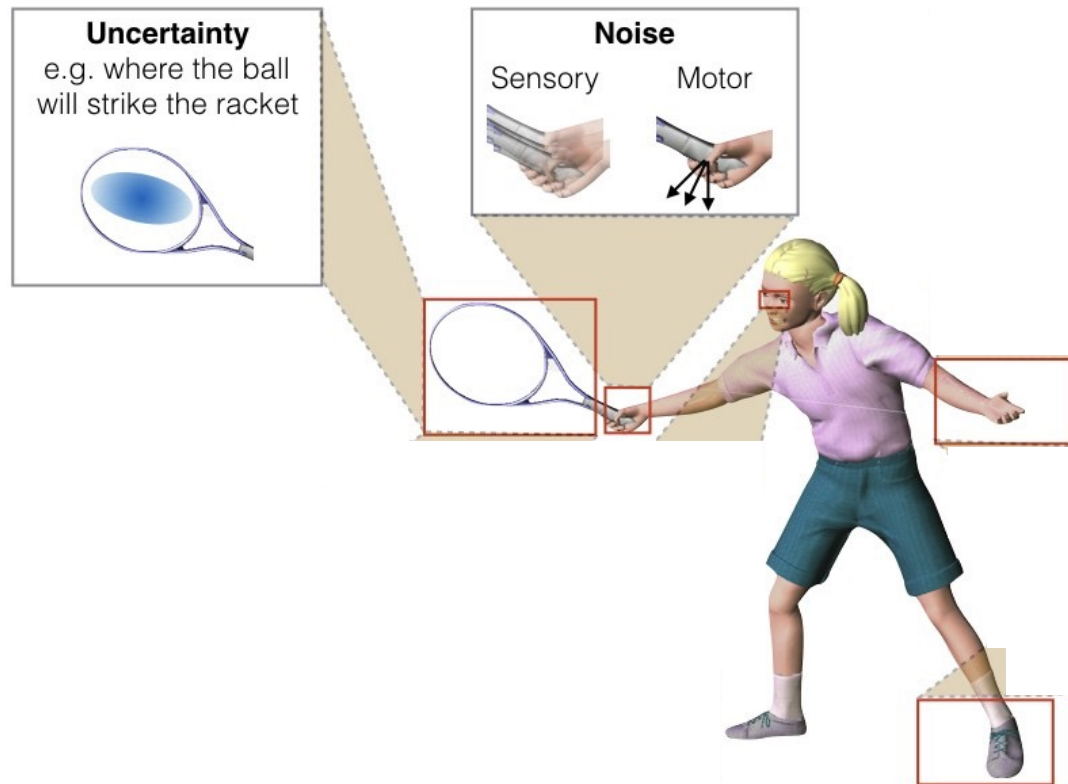


Motor noise is signal-dependent

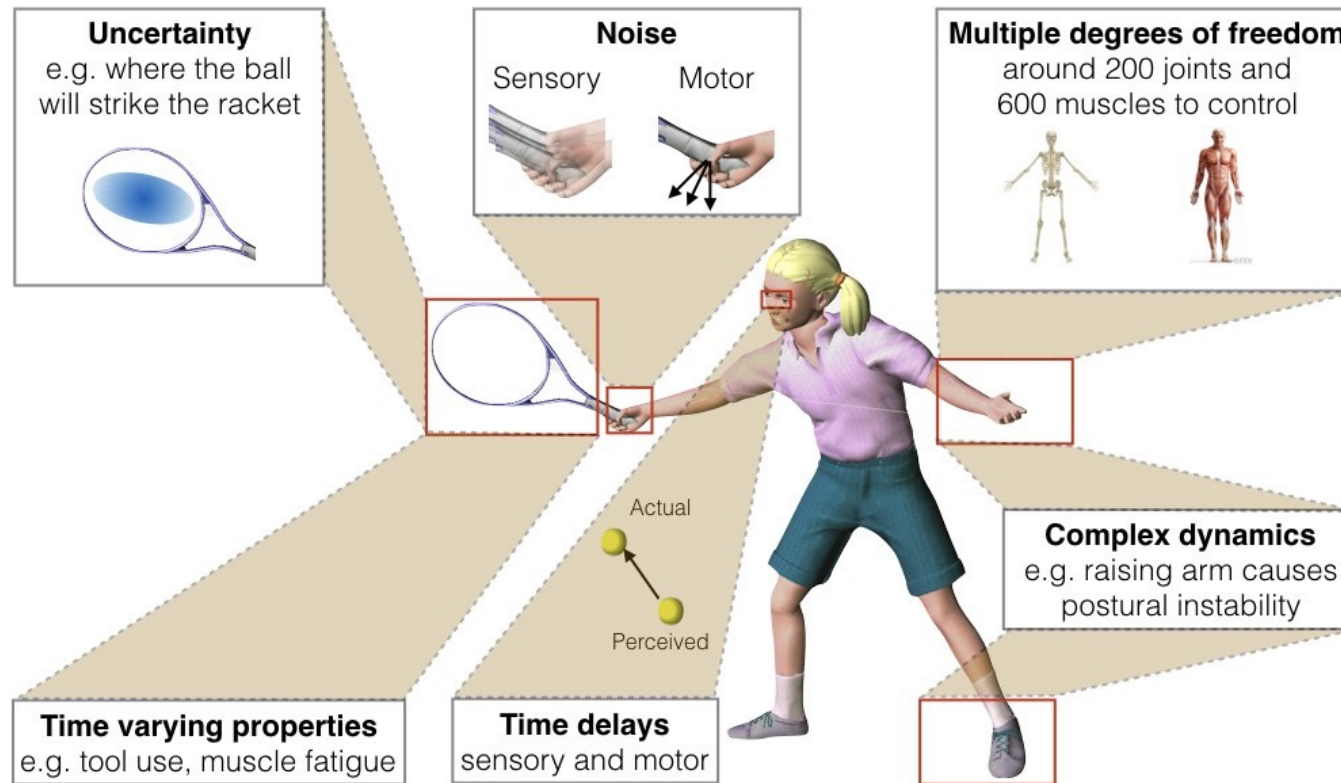


- The bigger the muscle group, the lower the slope (variability)
- For each muscle group, the more force you apply, the bigger the variability

Why is motor control so difficult?



Why is motor control so difficult?



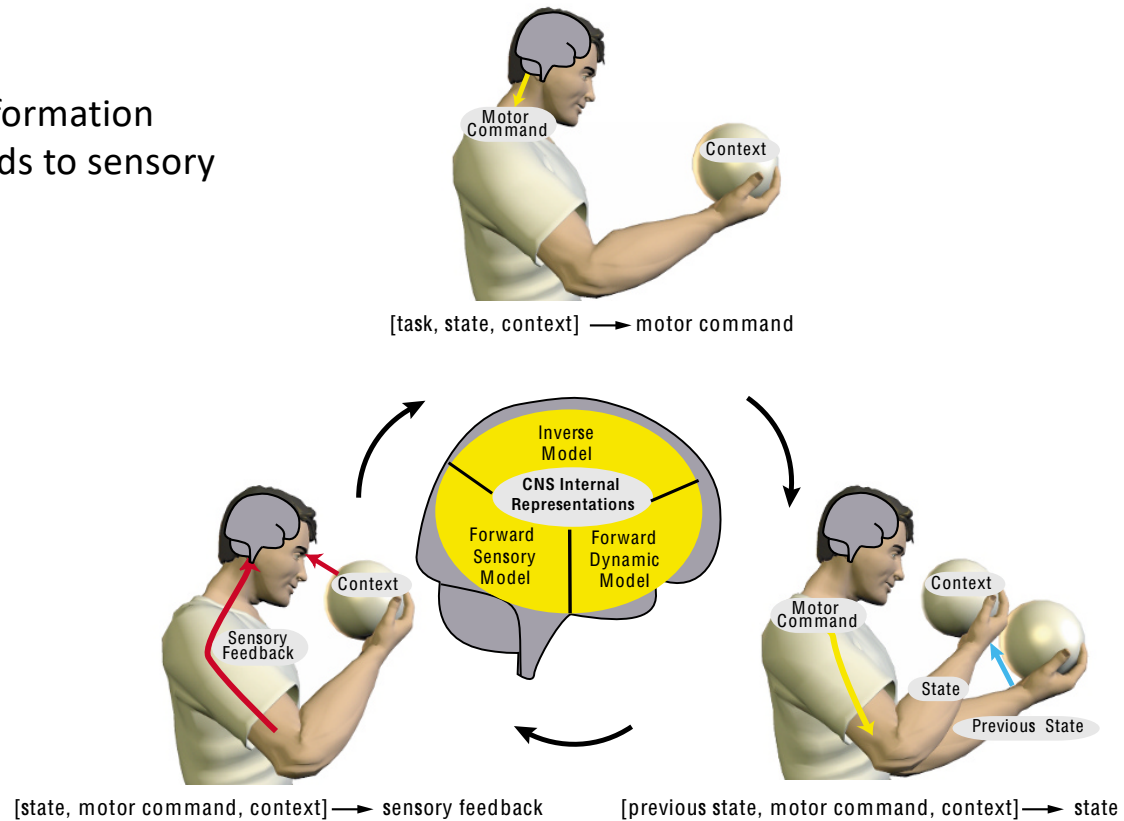
Principles of motor control

- need to bridge the gap from high level (abstract) goals and low level control (muscle activations)
- need to choose a specific action pathway out of infinite number of possibilities (redundancy)
- curse of dimensionality: given 600 muscles, 2^{600} possible muscle activations (each contracted or relaxed)
- how does the Central Nervous System deal with this?
 - represents the *State*, and *Context*

Internal models: the sensorimotor loop

Internal models:

Implement the transformation from motor commands to sensory signals and vice-versa



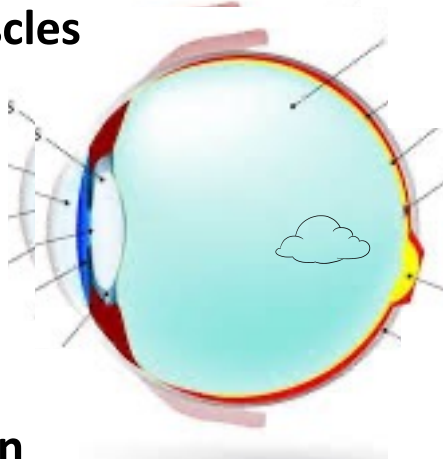
Forward models

- **What are they used for? (Why do we need them?)**
 - sensory feedback is slow and noisy
 - forward models compensate for sensorimotor delays (prediction)
 - estimating current and future states of the external environment and own body (before receiving sensory feedback), by using the **motor command**

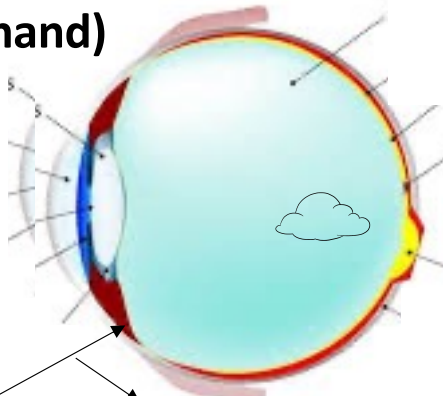


Helmholtz: how is eye position estimated?

**Sensors in
the eye muscles**



**Prediction
(motor command)**

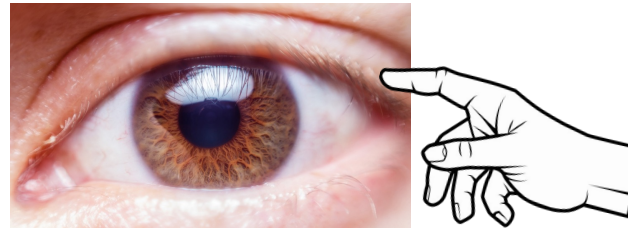


motor command

predictor

eye position

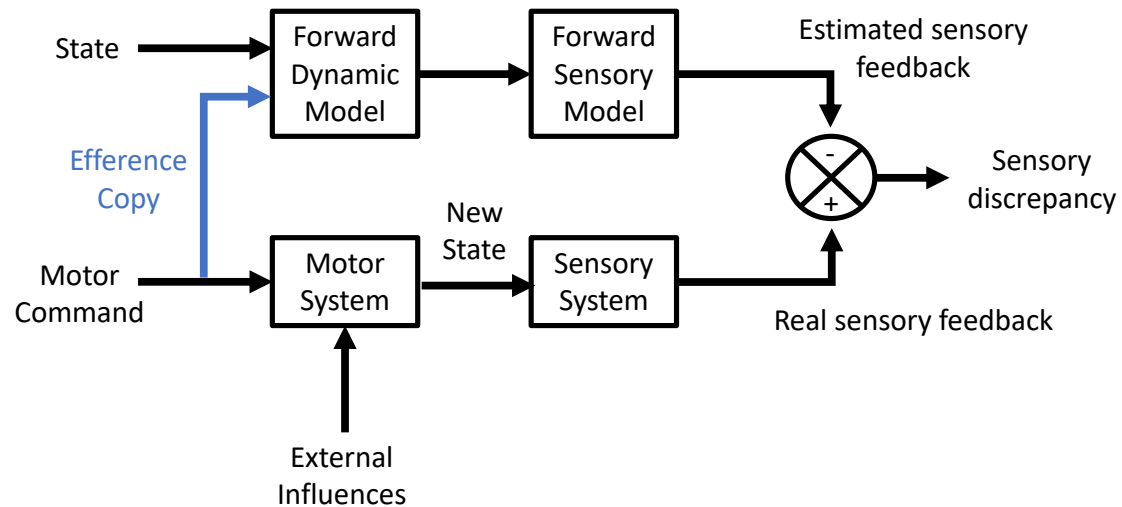
Is the world stable?



Does the world move?

Forward models

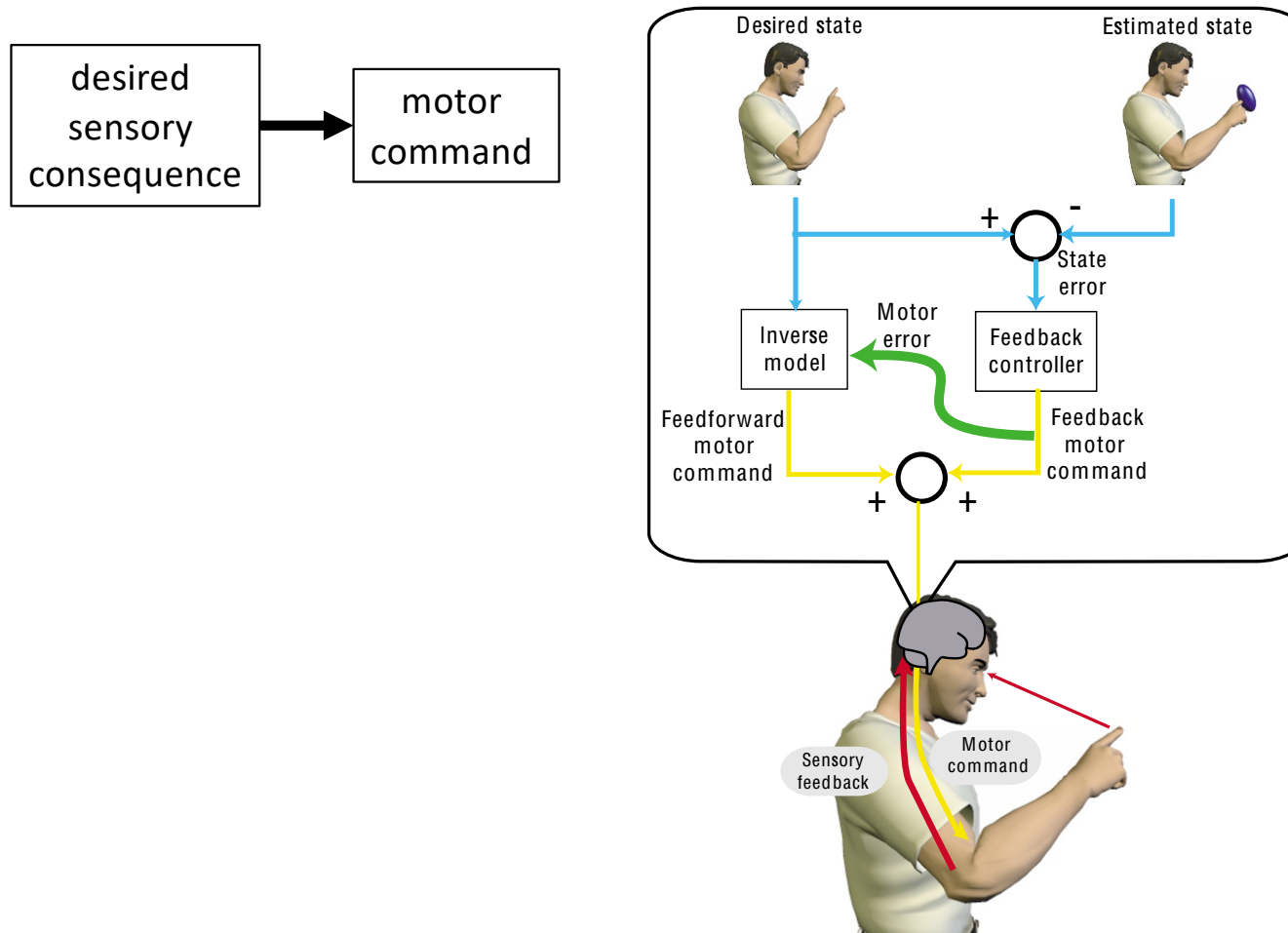
- **What are they used for? (Why do we need them?)**
 - predicting sensory consequences of actions
 - distinguish external influences from self-produced effects -> self-monitoring
 - increase sensitivity to unexpected events



Exercise

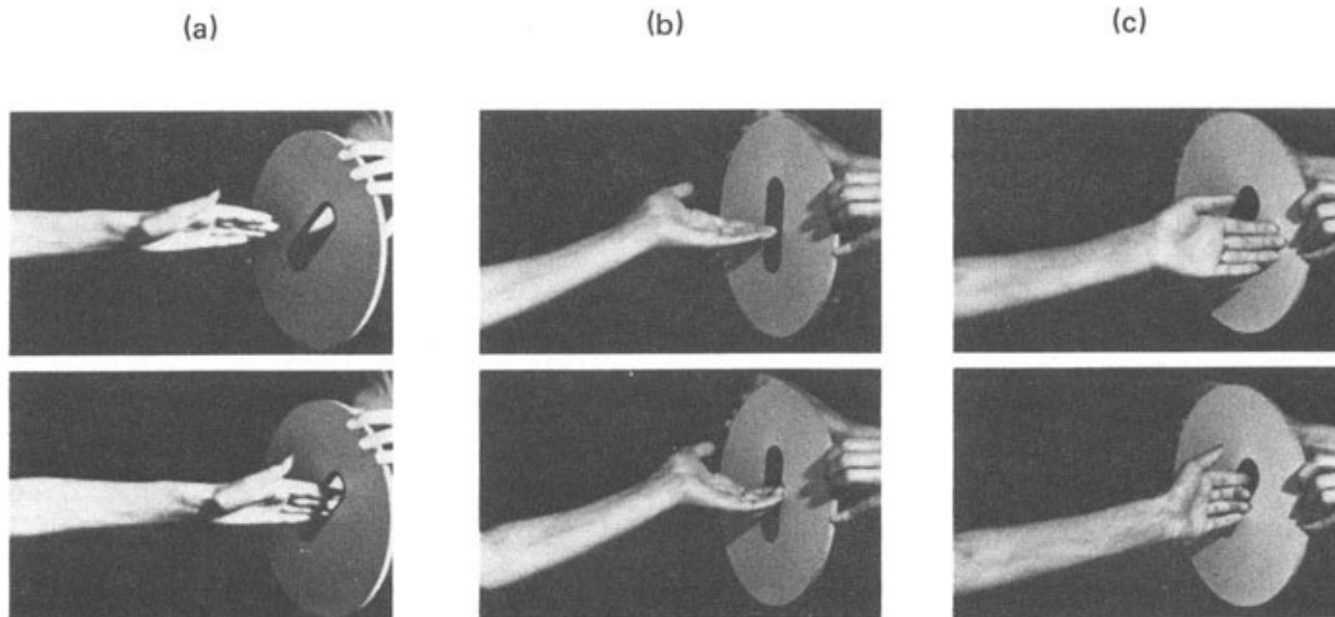
- With the person(s) next to you
 - discuss why you can't tickle yourself?
 - what is the purpose of this underlying mechanism?
 - what might be going wrong in your model if you could tickle yourself?
 - have you noticed that physical conflicts (with siblings or friends) tend to escalate? why might this be the case?
 - (15 minutes)

Inverse models – feedback-error learning model



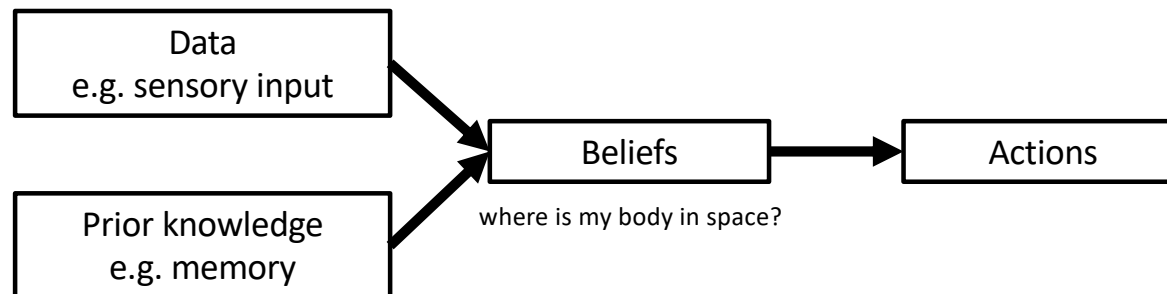
Optic ataxia

- inability to guide the hand toward an object using visual information

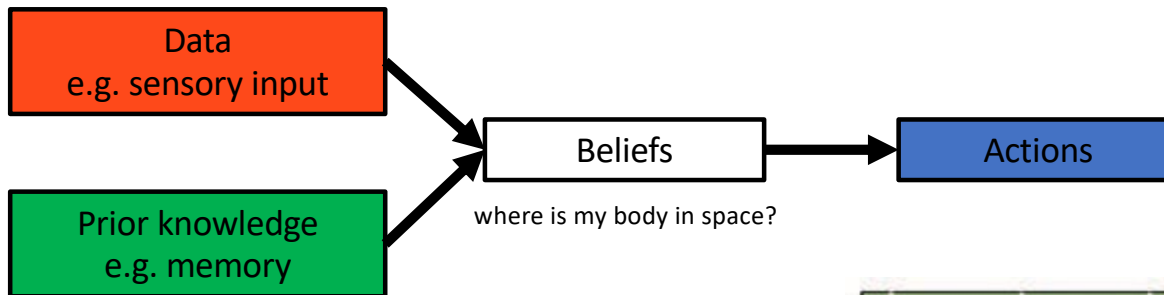


- patients are aware of their intentions, as well as having a problem with reaching
- inverse models not finely tuned by immediate context

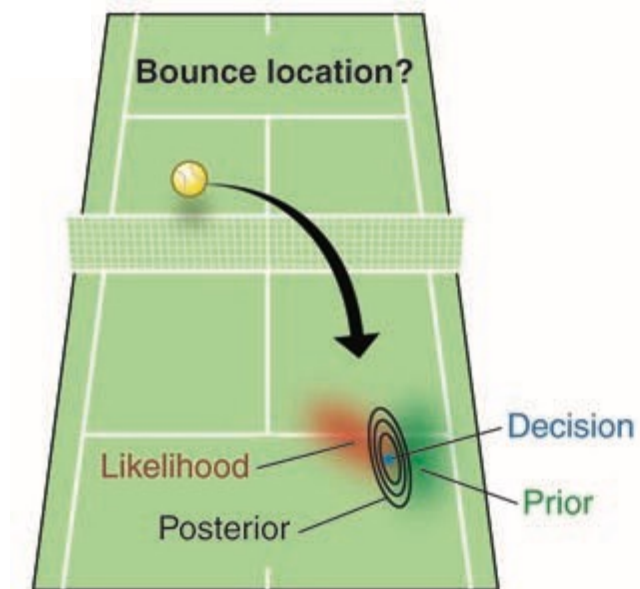
Bayes rule in action:
internal models embedded in probabilistic framework



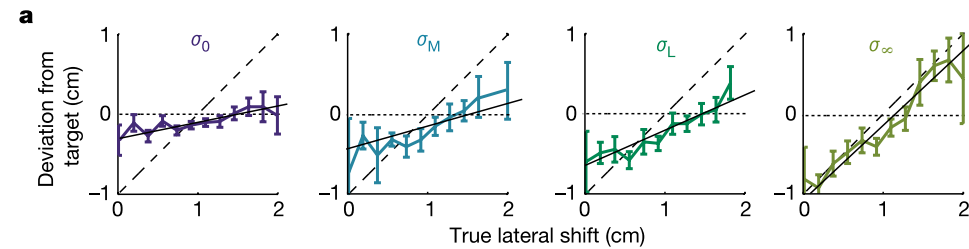
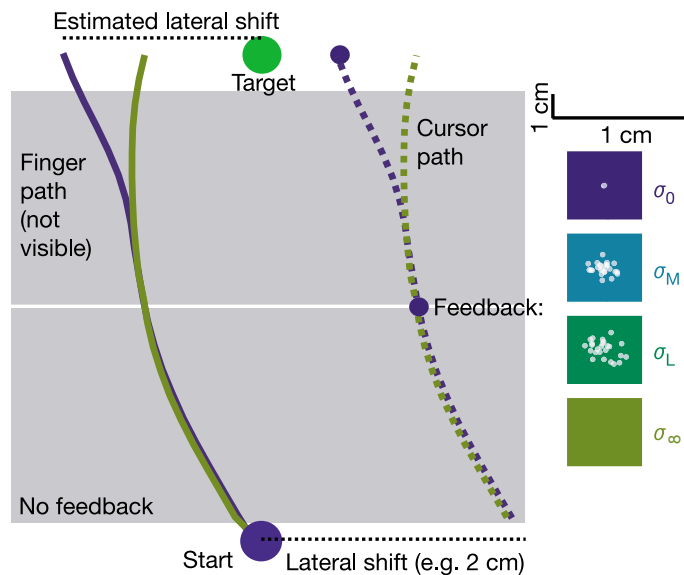
Bayes rule in action



$$\overbrace{P(\text{state}|\text{sensory input})}^{\text{Posterior}} = \frac{\overbrace{P(\text{sensory input}|\text{state})}^{\text{Likelihood}} \overbrace{P(\text{state})}^{\text{Prior}}}{P(\text{sensory input})}$$

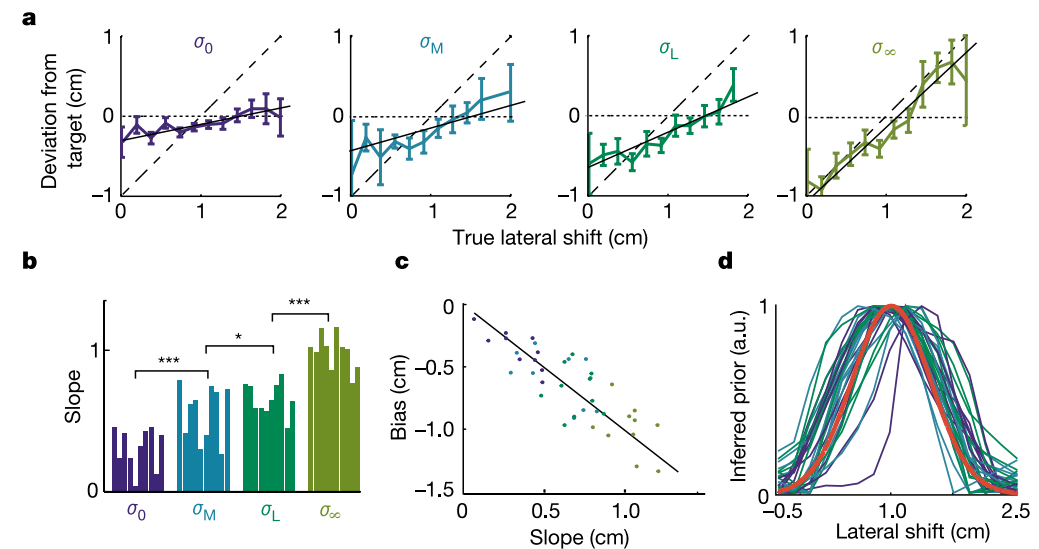
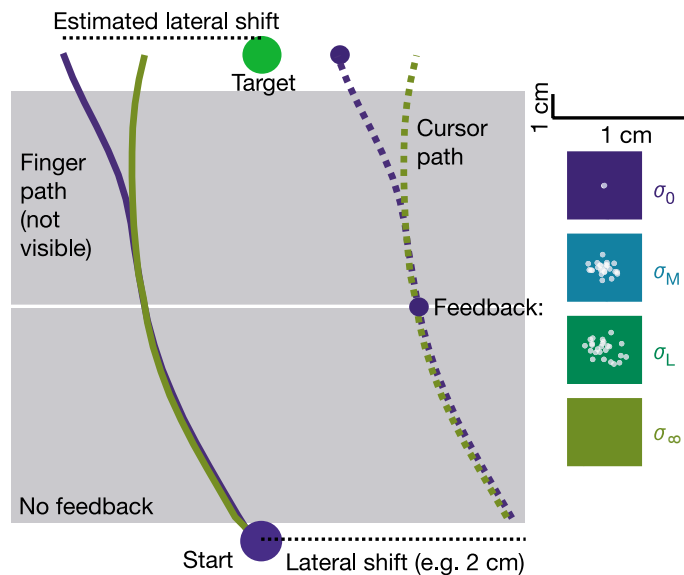


Evidence for Bayesian integration in sensorimotor learning



People optimally combine priors and likelihood as Bayesian estimators
When evidence is noisy (uncertain), people increase the weight of the prior

Evidence for Bayesian integration in sensorimotor learning



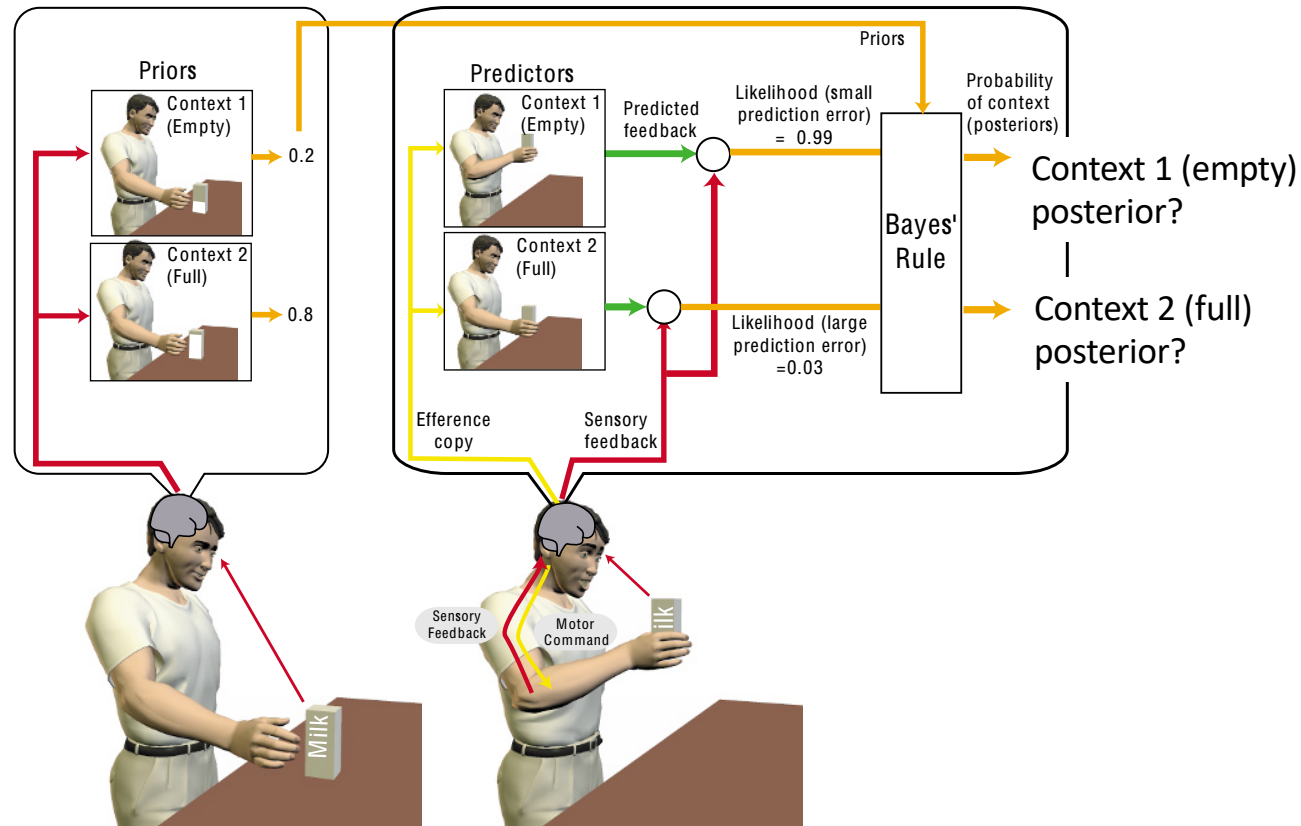
People optimally combine priors and likelihood as Bayesian estimators
 When evidence is noisy (uncertain), people increase the weight of the prior

Bayesian forward models (context estimation)

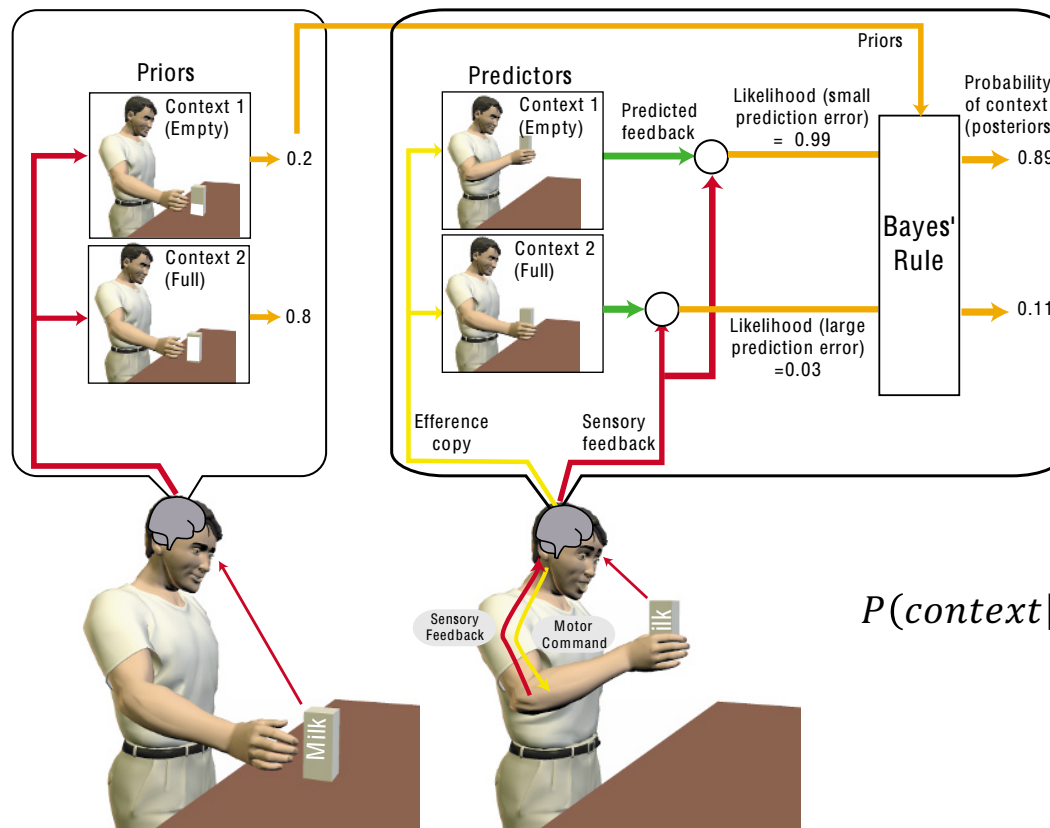
Predictive (forward) models in parallel

Hypothesis testers

Modular architecture can scale to 1000s of contexts



Bayesian forward models (context estimation)



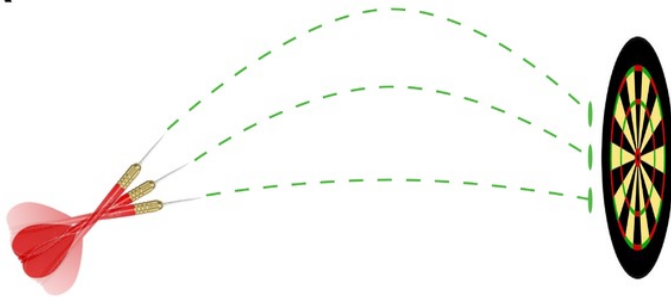
$$P(\text{context}|\text{data}) = \frac{P(\text{data}|\text{context})P(\text{context})}{\sum_{\text{contexts}} P(\text{data}|\text{context})P(\text{context})}$$

Motor control as decision making

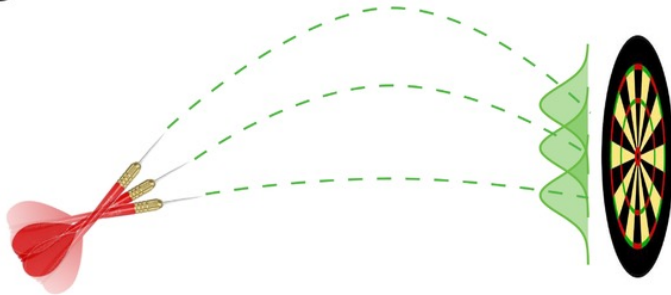
- how do we decide which action to take? or how to move our body? (what do we do with the posterior?)
- decision theory: problem of selecting the decision/action based on current belief
- goal: minimize the expected loss or cost (or maximize expected utility/reward) given our beliefs -> rational decision

Motor control as decision making

A



B



$$\sum_{\text{states}} L(\text{action}, \text{state}) P(\text{state} | \text{sensory input})$$

What is the “ideal” cost for goal-directed movements?

- it should make sense – evolutionary/learning advantage
- simple for brain to measure
- generalizes to different systems (eye, head, arm)
- generalizes to different tasks (pointing, grasping, drawing)
- reproduce and predict behaviour

What is the "ideal" cost for goal-directed movements?

- maximize accuracy
- minimize effort (energy)
- different tradeoffs depending on the type of movement (e.g. eating with knife and fork vs. walking)

Exercise

- 4 small problems: on internal models, decision making, and Bayes in action,
- similar to what you can expect on the exam